

Debugging C++

Mastering Error Identification



Syntax Fixes, Runtime Safety, and Logic Audits

COMPILE-TIME • RUNTIME • LOGICAL

1. The Gatekeeper: Compile-Time

Compile-time errors are mistakes that prevent the compiler from converting your C++ code into machine code. The program cannot even start.

"Fix me before you start."

These are usually mechanical issues—like using a language rule incorrectly.

```
int main() {  
    int x = 10  
    cout << x; // ERROR: Missing semicolon before 'cout'  
}
```

Common Causes:

- Missing semicolons (;) or brackets ({ }).
- Misspelled identifiers (Cout instead of cout).
- Declaring variables twice or using them without declaration.

2. The Execution Crash: Runtime

Runtime errors happen after your code has successfully compiled. The program starts, but it hits an impossible situation and "crashes."

"Something went wrong while running."

The OS often terminates the program to prevent damage to other system parts.

```
int main() {  
    int num = 100;  
    int divisor = 0;  
    cout << (num / divisor); // FATAL: Division by Zero  
}
```

The Infamous Segmentation Fault

This is a runtime error where your program tries to access memory that it doesn't own (like index 100 in an array of size 5).

3. The Silent Bug: Logical

Logical errors are the most dangerous. The program compiles and runs perfectly, but the output is **wrong**.

"Your program is thinking wrong."

The compiler can't help you here because you followed all the syntax rules; you just gave the wrong instructions.

```
float calculateArea(float r) {  
    return 2 * 3.14 * r; // LOGIC ERROR: This is Circumference, not Area!  
}
```

The computer did exactly what you told it to do—it just wasn't what you meant for it to do.

4. Mapping the Errors

Understanding where an error falls helps you choose the right tool to fix it.

Error Type	When It Appears	How to Find It
Syntax / Compile	During Compilation	Read Compiler logs
Runtime	During Execution	Use Debuggers/Logs
Logical	After Completion	Check output vs expectations

Quick Check: If your IDE underlines a line in red, it is a Compile-time error. If the console window vanishes suddenly, it is a Runtime error.

5. The Professional Toolset

Don't just stare at the code. Use these proven strategies to find the bugs:

1. The Print Method

Insert `cout` statements to see variable values at different stages. If `x` is correct on line 10 but wrong on line 20, you know exactly where to look.

2. Isolation

Use comments (`//` or `/* */`) to hide sections of code. If the error disappears when a section is hidden, the bug is in that section.

3. Read the Logs

C++ compiler messages can be long, but usually the **very first error** listed is the real cause. The rest are often "shadow errors" caused by the first one.

6. Debugging Checklist

- Check for missing `;` or mismatched `{ }`.
- Verify file existence before using `ifstream`.
- Ensure array indices stay within bounds.
- Validate user input before performing math.
- Verify formula logic manually with simple numbers.

The Debugger's Challenge

A CodeHive student wrote a function to calculate a 10% discount:

```
price = price * 0.10;
```

The program runs, but the student gets 10% of the price instead of the total price after discount.

Q: What type of error is this, and how do you fix it?

SYSTEM STABILIZED: ERROR HANDLING COMPLETE